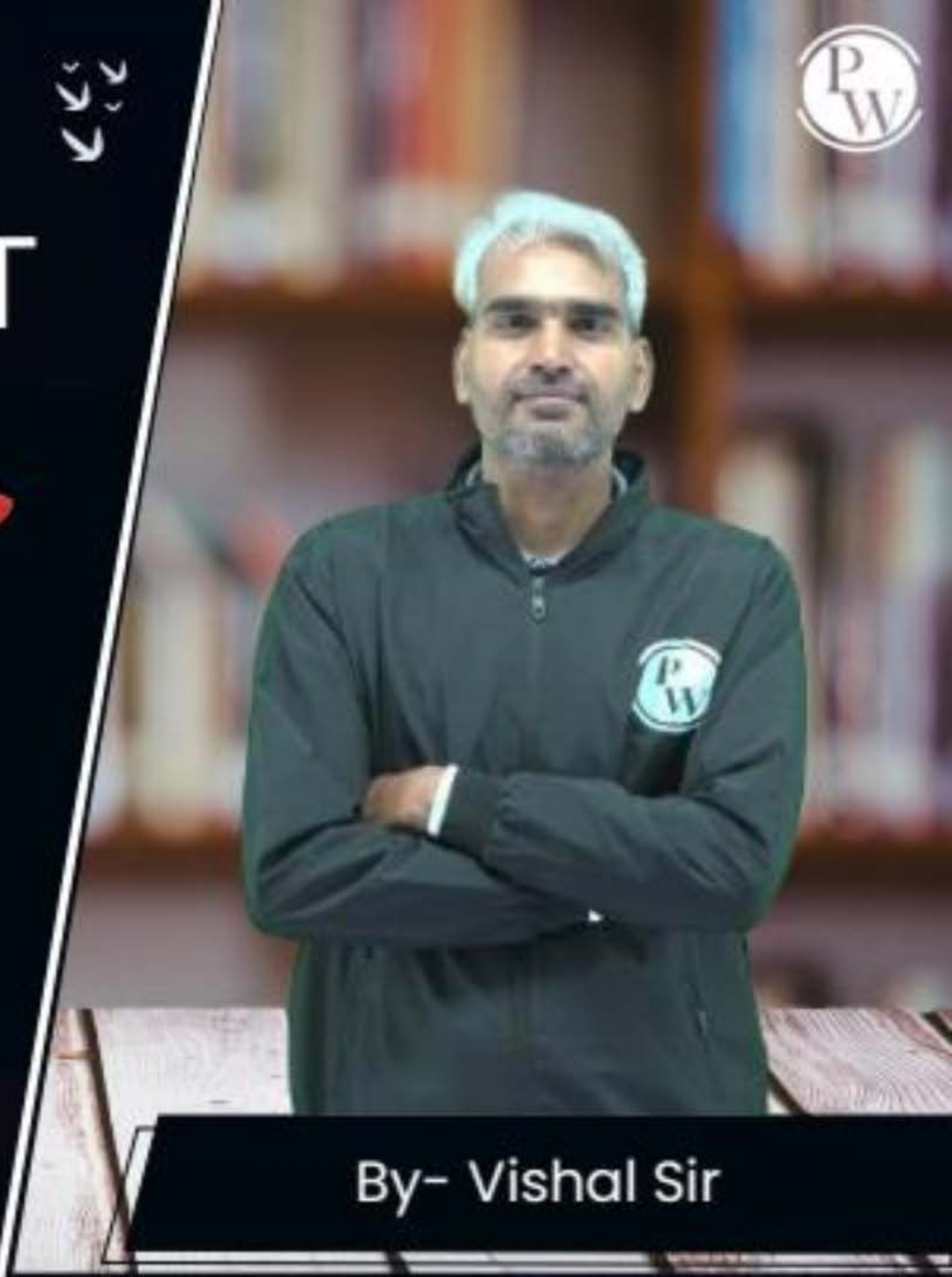
Computer Science & IT



Database Management System

Relational Model & Normal Forms

Lecture No. 17



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Recap of Previous Lecture



Topic Decomposition of relation up to BCNF

Topic Important properties

Slide

Topics to be Covered



- Topic: Multi-valued dependency
- Topic: Properties of Multi-valued dependency
- Topic: Fourth normal form (4NF)

Note:- ① Upto 3 NF we can always ensure lossless join decomposition as well as dependency Preserving decomposition

② While decomposing a relation into BCNF some times (Not always) it may not be possible to preserve some of the functional dependencies of the original relation, but we can always ensure lossless join decomposition.

③ Most adequate normal form of the database is 3NF, because we can always ensure lossless join decomposition as well as dep. preserving decomposition {i.e. No loss of information}

✓ ④ In a relation if all attributes are prime attributes, then the relation is at least in 3NF.

✓ ⑤ If all candidate keys of the relation are simple candidate keys, then partial dependency can not exist in the relation, therefore relation will be at least in 2NF.

✓ ⑥ In a relation if all attributes are prime attributes, and all candidate keys of the relation are simple candidate keys, then relation is always in BCNF

⑦ A binary relation {i.e. a relation with exactly two attributes} is always in BCNF.

Let R(AB) is a relation with two attributes, then only four different types of FD set are possible

$F_1 = \{A \rightarrow B\}$ (BCNF)

Ck : A

$F_2 = \{B \rightarrow A\}$ BCNF

Ck : B

$F_3 = \{A \rightarrow B$ (BCNF), $B \rightarrow A$ (BCNF)$\}$

Ck : (A), (B)

$F_4 = \{\}$ Empty

Ck : (AB) & Rel^n is in BCNF

⑧ If a relation is in BCNF, then there will be no redundancy in that relation because of functional dependencies,

but redundancy may still be present in the relation because of multi-valued dependencies.

eg:

Sid	Cid	Mob_No
S_1	C_1	M_1
S_1	C_1	M_2
S_1	C_2	M_1
S_1	C_2	M_2
S_2	C_2	M_2

Redundancy exists, but it is not because of Functional dependencies. it is because of M.V.D.

- $Sid \not\rightarrow Cid$
- $Sid \not\rightarrow Mob_ND$
- $Cid \not\rightarrow Sid$
- $Cid \not\rightarrow Mob_ND$
- $Mob_ND \not\rightarrow Sid$
- $Mob_ND \not\rightarrow Cid$
- $Sid, Cid \not\rightarrow Mob_ND$
- $Sid, Mob_ND \not\rightarrow Cid$
- $Cid, Mob_ND \not\rightarrow Sid$

None of the non-trivial functional dependency exists in the given relation.

∴ Given relation is in BCNF



Topic : Multivalued dependency

"X multi valued determines Y" is denoted by $X \twoheadrightarrow Y$

- If there exist two or more independent attributes which are dependent on some other set of attributes then multi-valued dependency will(may) exist in the relation

★ Formal definition of MVD :

Let R be the relation, and X & Y are some set of attributes from relation R,

{ let us define $Z = \text{Attributes}_{R} - (X \cup Y)$

i.e; Z is a set of remaining attributes of R }

If there exists at least 4 tuples $t_1, t_2, t_3, t_4 \in R$ such that:

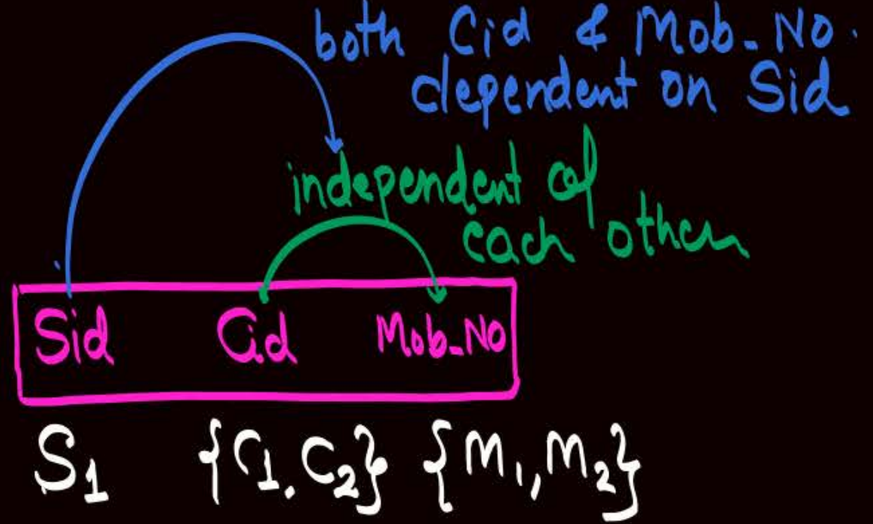
- $t_1.X = t_2.X = t_3.X = t_4.X$ and
- $t_1.Y = t_2.Y$ and $t_3.Y = t_4.Y$ and
- $t_1.Z = t_3.Z$ and $t_2.Z = t_4.Z$

then Multi-valued dependency $X \rightarrow\rightarrow Y$ exists in R.

X Z Y ⟹ w.r.t. this new ordering Sid →→ Mob_No also exists in the relation R

X Y Z

		Sid	Cid	Mob_No
t_1	t_1	S_1	C_1	M_1
t_3	t_2	S_1	C_1	M_2
t_2	t_3	S_1	C_2	M_1
t_4	t_4	S_1	C_2	M_2
		S_2	C_2	M_2



Let X = Sid & Y = Cid, then Z = Mob_No.

We can observe,

$t_1.X = t_2.X = t_3.X = t_4.X$

$t_1.Y = t_2.Y$ (and) $t_3.Y = t_4.Y$

$t_1.Z = t_3.Z$ (and) $t_2.Z = t_4.Z$

$\therefore X \twoheadrightarrow Y$

i.e. Sid →→ Cid exists in R

* Another definition w.r.t. MVD :-

Whenever we swap the values of attribute set Y in two tuples {let t_1 & t_2} which agree on the value of attribute set X {i.e., $t_1.X = t_2.X$}, if the resulting tuple was already a member of the relation, then $X \rightarrow\rightarrow Y$ exists in the relation.

Sid	Cid	Mob_No
S_1	C_1	M_1
S_1	C_1	M_2
S_1	C_2	M_1
S_1	C_2	M_2
S_2	C_2	M_2

- Check whether Cid →→ Mob_No exists in the relation or not?

S_1 C_2 M_2

S_2 C_2 M_1

Not already present in the relation

∴ Cid →→ Mob_No does not hold in the relation



Topic : Properties of Multivalued dependency

(1) **Complementation Rule :-** Let R be the relational schema and X & Y be some set of attributes over relation R,

If $X \twoheadrightarrow Y$ exist in relation R.

then $X \twoheadrightarrow (\text{Attributes of } R - (X \cup Y))$ also exist in relation R

let $Z = \text{Attributes of } R - (X \cup Y)$

i.e. $X \twoheadrightarrow Z$ will also exist in the relation



Topic : Properties of Multivalued dependency

② Trivial MVD :-

Let $X \rightarrow\rightarrow Y$ exist in relation R,

$X \rightarrow\rightarrow Y$ is called a trivial MVD

if $X \supseteq Y$

or

$X \cup Y = R$ { i.e; Attributes of X ∪ Attributes of Y = Attributes of R }



Topic : Properties of Multivalued dependency

③ Transitivity :-

If $X \rightarrow\rightarrow Y$ and $Y \rightarrow\rightarrow W$ exists in relation R

then $X \rightarrow\rightarrow (W - Y)$ will also exist in the relation.



Topic : Properties of Multivalued dependency

④ Augmentation :

If $X \rightarrow\rightarrow Y$ exist in the relation R, and α, β are two set of attributes from relation R, such that $\alpha \supseteq \beta$.

then $X\alpha \rightarrow\rightarrow Y\beta$ will also hold true in relation R.



Topic : Properties of Multivalued dependency

⑤ Replication :-

- Every Functional dependency is also a M.V.D.
  { Every MVD need not be a Functional dependency }

- i.e. if $X \rightarrow Y$ exist in R, then $X \rightarrow\rightarrow Y$ will also exist in R
- & if $X \rightarrow\rightarrow Y$ exist in R, then $X \rightarrow Y$ need not exist in R.



Topic : Properties of Multivalued dependency

⑥ Splitting

Multi-valued dependencies are not allowed to split.

i.e. if $X \rightarrow\rightarrow YW$ exist in relation, then, $\begin{cases} X \rightarrow\rightarrow Y \text{ and } X \rightarrow\rightarrow W \\ \text{need not hold true.} \end{cases}$

But FDs are allowed to split.

i.e. if $X \rightarrow YW$, then $X \rightarrow Y$ & $X \rightarrow W$



Topic : Fourth normal form (4NF)

* For a relation R to be in 4 NF:

① Relation R must be in BCNF.

and ② Every non-trivial MVD $X \twoheadrightarrow Y$.

must be with X as a Super key of the relation

{ i.e. Every non-trivial FD $X \rightarrow Y$ must be with 'X' as a Super key }

Note:- Superkeys and Candidate keys are always identified by non-trivial FDs; because FDs are Key dependencies whereas MVDs are Data dependencies.

Sid	Cid	Mob_No
S_1	C_1	M_1
S_1	C_1	M_2
S_1	C_2	M_1
S_1	C_2	M_2
S_2	C_2	M_2

No non-trivial FD exists in the relation,
∴ Relation is in BCNF,
and Candidate key of the relation is = {Sid, Cid, Mob_No.}

Following Non-trivia MVDs exists in the relation

(i) Sid $\rightarrow\!\!\rightarrow$ Cid
Not a SK

(ii) Sid $\rightarrow\!\!\rightarrow$ Mob_NO.
Not a SK

Non-trivial MVDs of type $X \rightarrow\!\!\rightarrow Y$ exists in the relation, such that 'X' is not a Superkey
∴ Relation is not in 4NF.

* If the relation is in BCNF, but not in 4NF,
then, for 4NF decompose the relation w.r.t.
Every non-trivial MVD $X \rightarrow\rightarrow Y$, in which X is not a S.k.

Sid	Cid	Mob_No
S_1	C_1	M_1
S_1	C_1	M_2
S_1	C_2	M_1
S_1	C_2	M_2
S_2	C_2	M_2

No non-trivial FD exists in the relation, ∴ Relation is in BCNF, and Candidate key of the relation is = (Sid, Cid, Mob_No.)

Following Non-trivia MVDs exists in the relation

(i) Sid →→ Cid
Not a SK

(ii) Sid →→ Mob_NO.
Not a SK

← Not in 4NF

⇓

For 4NF decompose w.r.t non-trivial MVDs in which LHS is not a S.K.

Sid →→ Cid

Sid	Cid
S_1	C_1
S_1	C_2
S_2	C_2

No non trivial FD, ∴ BCNF
Sid →→ Cid exists in relation but X∪Y = R, ∴ it is trivial
∴ No non-trivial MVD ∴ 4NF

Sid →→ Mob_NO

Sid	Mob_No.
S_1	M_1
S_1	M_2
S_2	M_2

No non trivial FD, ∴ BCNF
Sid →→ Mob_NO. exists in relation but X∪Y = R, ∴ it is trivial
∴ No non-trivial MVD ∴ 4NF



2 mins Summary

- **Topic** Multi-valued dependency
- **Topic** Properties of Multi-valued dependency
- **Topic** Fourth normal form (4NF)

Slide

THANK - YOU